

POCKET PET AI

A frozen-model personal operating system

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Pocket Pet AI™

A frozen-model wearable assistant with personality

Core concept:
ear wearable + pocket compute puck + phone control

- Model:** 7B–9B frozen low-bit omni assistant
- Weights:** ~2–4 GB frozen model package
- Memory:** 8–16 GB RAM
- Context:** 32k typical, up to 120k with compressed latent KV
- Power:** 8–15W typical, 20–30W burst
- Battery:** 30–60 Wh
- Runtime:** 2–6 hours active use
- Connectivity:** Bluetooth, Wi-Fi, USB-C
- Features:** real-time voice, personality, phone control, private on-device memory

POCKET-SIZED POWER
Premium matte finish. Soft-touch, durable, and made to go where you go.

USB-C CHARGING
Fast charging and data. Universal and future-proof.

1 Always-listening low-power mode
Intelligent wake word detection with ultra-low power standby.

2 Real-time speech + TTS
Natural conversations with low-latency voice responses.

2 Real-time speech + TTS
More context, less RAM. Up to 120k with compressed latent KV.

3 Compressed KV / latent memory
More context, less RAM. Up to 120k with voice responses.

4 Control your phone through a companion app
Manage settings, permissions, and actions—hands-free or at a glance.

REMOVABLE BATTERY
Hot-swappable battery module. Extend your runtime in seconds.

EAR WEARABLE
Lightweight open-ear clip-on design. Stay aware. Stay free.

PHONE CONTROL
Your AI assistant. Your data. Your rules. Full control in your hand.



The image shows the Pocket Pet AI hardware and its companion app. The hardware consists of a black, rounded rectangular 'pocket compute puck' with a small 'P' logo on top and a USB-C port on the bottom. Above the puck are two black, over-ear headphones with a small 'P' logo on the side. To the right is a smartphone displaying the 'Pocket Pet AI' app. The app interface includes a 'Connected' status bar, a 'Tap to talk' button, and a 'Permissions & Controls' section with toggles for Microphone, Location, Camera, and Contacts. At the bottom of the app are icons for Home, Messages, Activity, and Settings. The smartphone screen also shows a 'Phone Control' section with a 'Manage' button.

TECHNICAL WHITEPAPER / OPEN RESEARCH DRAFT / JULY 2026

From real-time software baseline to an instrumented ternary FPGA prototype

This document is a feasibility program, not a performance claim. It turns the product vision into equations, measurable budgets, falsifiable experiments and staged decision gates.

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Executive thesis

What is proposed, what is known, and what must be proven

THE PROPOSITION

A stable brain, optimized as a complete system

Pocket Pet AI combines an open-ear interface, a pocket compute puck, and permissioned phone and desktop companions. Its defining constraint is that the deployed foundation model is frozen. That constraint can convert a general inference problem into a co-designed model, memory layout, decoder and accelerator - while writable personal memory and tool adapters evolve independently.

The near-term target is a local, streaming voice agent whose response begins fast enough to feel conversational and whose sensitive actions pass through deterministic policy enforcement. The long-term hypothesis is that a natively ternary 7B-9B model, fixed layout, compressed KV architecture and model-specific hardware can fit a 2-4 GB model package and operate within a pocketable power envelope.

The hypothesis is plausible, not established. BitNet b1.58 reports competitive results for ternary models trained from scratch, but it is explicitly a work in progress and does not demonstrate this multimodal product, this quality target, this power target, or an FPGA implementation [1]. The correct engineering response is staged evidence, not extrapolation.

CLAIM DISCIPLINE

Label	Meaning	Example in this paper
Published	Reported by a cited source	Ternary weights in {-1,0,+1} [1]
Derived	Arithmetic from declared assumptions	$9B \times \log_2(3) = 1.78 \text{ GB ideal entropy}$
Target	A design objective, not measured	<450 ms median time to first audio
Hypothesis	Requires experiment	Fixed weight banking reduces effective weight traffic

The product succeeds only if all four layers close together: model quality, memory traffic, interactive latency, and permission safety. A token-per-second headline alone is insufficient.

DECISION SUMMARY

Decision	Current recommendation	Reason
Stage 1 model	Start with 1B-3B local candidates, then scale	Faster iteration and measurable baseline before 7B-9B
Attention	GQA baseline; latent KV as a trained variant	KV architecture is not a post-hoc implementation detail
Acceleration	FPGA kernel slice before full-model mapping	Proves packing, bandwidth and timing with bounded scope
Phone control	Android-first, typed actions, confirmation policy	Accessibility is powerful and sensitive; iOS is more constrained
ASIC	Do not commit before Stage 2 gates	Model and decoder must be frozen before silicon

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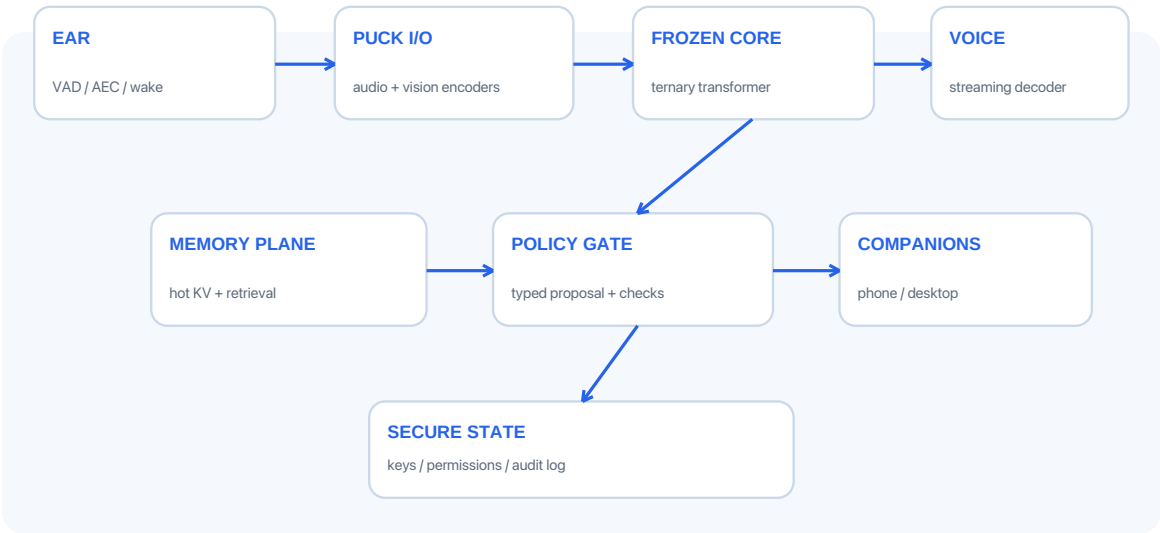
System architecture

The wearable is I/O; the puck is cognition; companions are controlled hands

REFERENCE SYSTEM

One assistant, four trust zones

Trust boundaries are explicit: the model proposes; deterministic software authorizes and executes.



The model never receives an unrestricted device handle. It emits a typed proposal such as `message . compose ( contact_id , body )`. A policy service validates schema, capability, user presence, destination and confirmation class. A companion executes the approved primitive, then returns an observation. This proposal-check-execute-observe loop is part of correctness, not a UI accessory.

AUDIO PATH

Block	Mode	State	Budget hypothesis
VAD / wake / AEC	Always on	DSP local	10-100 mW class; measure on target
Speech encoder	Streaming after VAD	Puck accelerator	80 ms frame cadence; causal chunks
Core model	Prefill then decode	Frozen engine	TTFT dominates first response
Speech decoder	Streaming	Dedicated small model	Start after semantic buffer is safe
Ear link	Duplex audio + control	Wireless	Jitter buffer must remain bounded

PERSONAL MEMORY

The KV cache is a computational cache, not a life history. Long-term memory is an encrypted, user-inspectable store of events, facts, preferences and summaries. Retrieval inserts a small evidence bundle into the active context. Deletion must remove both the structured record and searchable derived artifacts. The frozen model stays frozen; relationship state remains writable and revocable.

Multimodality is event-driven: a tiny classifier or explicit user request activates image capture; selected frames become compact visual tokens; raw pixels are discarded unless retention is requested. Continuous high-resolution video is outside the nominal thermal envelope.



Model and numerical design

Native ternary training, protected islands, and a reproducible package

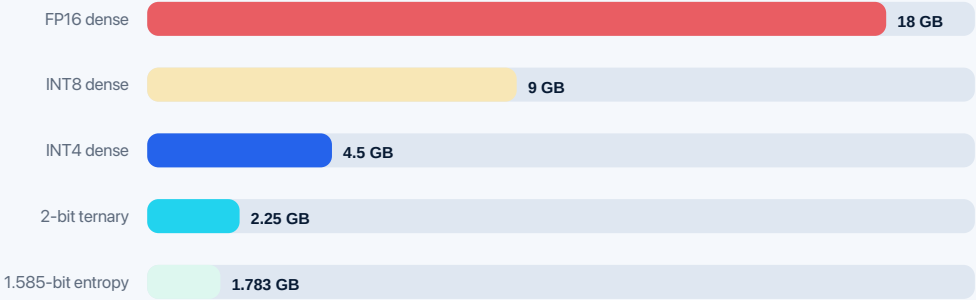
TERNARY CORE

1.58 bits is an information bound, not a file format

For ordinary weights w in $\{-1,0,+1\}$, the ideal code length is $\log_2(3) = 1.585$ bits per symbol under a uniform distribution. Real hardware commonly uses two physical bits, bitplanes plus metadata, or entropy-coded blocks. Compute is conceptually $y = \alpha * \sum_i q_i x_i$, where q_i is ternary and α is a learned or derived scale. Products reduce to select, add, subtract or skip, but accumulation, activation quantization, normalization and scaling remain real work.

Ideal dense payload: $M_{ideal} = N \log_2(3) / 8$. Physical two-bit payload: $M_{2b} = 2N / 8$. For $N = 9e9$: 1.78 GB decimal ideal and 2.25 GB decimal at two bits. A package must additionally carry embeddings, output head, scales, norms, tokenizer, modality towers, alignment padding, checksums and any protected weights.

9B language-core storage scenarios (decimal GB, derived)



MIXED-PRECISION POLICY

Parameter class	Candidate representation	Why	Acceptance test
Linear weights	Ternary + block scale	Primary density and add/sub path	Per-layer SQNR and task delta
Embeddings / LM head	4-8 bit or tied	High reuse; often quality sensitive	Rare token and multilingual tests
Norm / scale	FP16 or BF16	Small footprint; numerically sensitive	Drift over long decode
Outliers / protected set	4-8 bit sparse sidecar	AWQ finds salient weights via activations [6]	Ablate sidecar density
Activations	8 bit baseline; lower experimental	Dynamic range changes by token	Calibration and saturation counters
Accumulators	INT24/32 or block FP	Avoid overflow and excessive rounding	Bit-exact stress vectors

Training from scratch is the evidence-backed path for BitNet-style behavior [1]. Post-training conversion of an arbitrary model to ternary should be treated as a separate, higher-risk experiment. The prototype must record exact checkpoint, tokenizer, calibration set, quantizer version, seeds, package hash and evaluation commit.



Memory and bandwidth

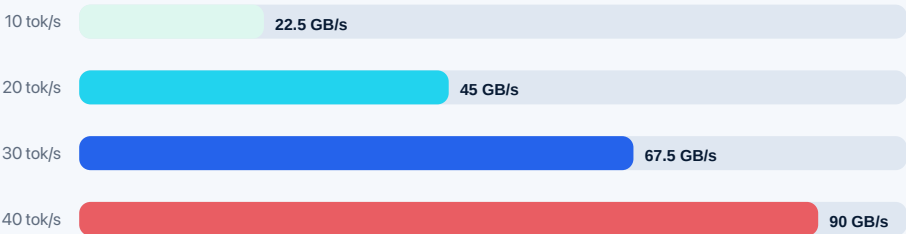
Weights fit; movement determines whether the system performs

LOWER BOUNDS

Decode is a bandwidth problem until proven otherwise

If every weight is streamed once per generated token, the memory-bandwidth lower bound is $B_{\text{weight}} = \text{package_bytes} \times \text{tokens_per_second}$. A 2.25 GB two-bit 9B core at 20 tok/s asks for 45 GB/s before KV traffic, activations, modality blocks or inefficiency. At 40 tok/s it asks for 90 GB/s. On-chip reuse across a single batch-one token is limited; fixed placement helps addressing and compute, but does not make data movement free.

Weight-stream bandwidth floor for a 2.25 GB core



KV CACHE EQUATION

For a conventional decoder cache, bytes approximately equal $L \times T \times 2 \times H_{\text{kv}} \times D_{\text{h}} \times b$, where L is layers, T tokens, the factor 2 stores K and V , H_{kv} is KV heads, D_{h} is head dimension, and b is bytes per element. For an illustrative 32-layer GQA model with 8 KV heads, head dimension 128, and 2-byte values, this is 128 KiB per token: 4.0 GiB at 32K and 15.0 GiB at 120K. At one byte it halves. This is why attention architecture is a product-level choice.

Illustrative cache	Bytes / token	32K context	120K context	Status
MHA: L32, Hkv32, Dh128, FP16	512 KiB	16.0 GiB	60.0 GiB	Derived
GQA: L32, Hkv8, Dh128, FP16	128 KiB	4.0 GiB	15.0 GiB	Derived
GQA: same, INT8	64 KiB	2.0 GiB	7.5 GiB	Derived
Latent: L32, Dlatent512, INT8	16 KiB	0.5 GiB	1.88 GiB	Illustrative
Target hot+cold hierarchy	variable	0.06-0.25 GB	0.25-1.2 GB	Target; must train

GQA uses fewer KV heads and can approach multi-head quality with multi-query-like speed [3]. DeepSeek-V2 reports a 93.3% KV-cache reduction from Multi-head Latent Attention in its own architecture [2]; that result does not automatically transfer to a Pocket Pet model. FlashAttention demonstrates that tiling can reduce off-chip reads and writes for exact attention [4]. These are candidate ingredients, not stackable percentage guarantees.

MEMORY HIERARCHY

Tier	Capacity hypothesis	Contents	Design rule
Register / local SRAM	sub-MB to few MB	lane operands, partial sums	No avoidable spills
Shared on-chip SRAM	32-256 MB aspirational	tiles, hot activations, buffers	Double-buffer and bank explicitly
LPDDR	8-16 GB product target	KV, OS, retrieval, temporary vision	Page and instrument bytes moved
Frozen store	2-4 GB package target	weights, scales, manifests	Integrity checked; deterministic layout
Encrypted flash	32-128 GB	memories, logs, updates	Keys hardware-bound; retention controls



Latency, energy and thermals

User-perceived response is an end-to-end critical path

INTERACTION BUDGET

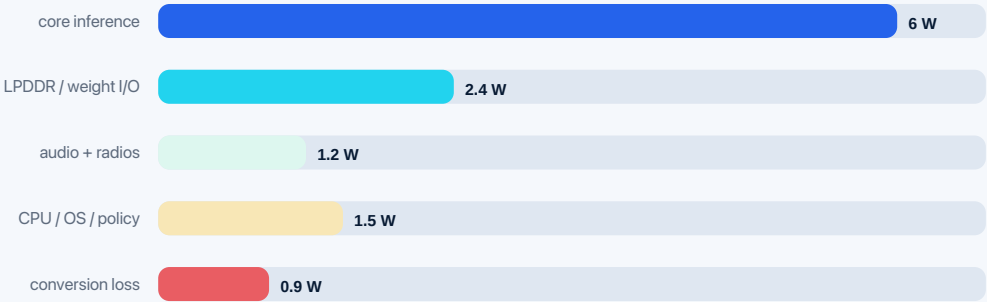
Optimize time to first audio, not just peak tokens per second

A credible voice metric begins at the end of detected user speech and ends at the first intelligible synthesized audio. It includes endpointing, final encoder frames, retrieval, prompt assembly, prefill, first semantic tokens, speech buffering, vocoder startup and radio jitter. Report P50, P95 and P99 separately, with prompt-length buckets.

Stage	Stage 1 target	Measurement boundary	Failure symptom
Endpoint detection	80-180 ms	last speech energy to final endpoint	assistant feels hesitant
Retrieval + policy context	<40 ms P50	query ready to context ready	long-tail stalls
Prefill + first token	<180 ms at nominal prompt	context ready to first token	context length dominates
Semantic buffer + vocoder	<100 ms	first token to audio PCM	speech starts late or glitches
Link / jitter	<50 ms	PCM ready to ear output	dropouts
Composite	<450 ms P50; <800 ms P95	end of speech to first audio	not conversational

These are proposed gates, not measured performance. Endpointing and the semantic buffer can overlap some computation, so component medians do not simply sum; the trace must preserve causality.

Illustrative active power allocation at 12 W nominal



ENERGY MODEL

Energy per turn is $\int P(t) dt$. Battery runtime is not $\text{battery_Wh} / \text{peak_W}$; it depends on duty cycle, conversion efficiency, standby and burst duration. An example mixed hour with 12 minutes at 12 W, 6 minutes at 5 W and 42 minutes at 0.7 W consumes 3.19 Wh. A usable 45 Wh pack at 90% system efficiency would therefore support roughly 12.7 such hours. This is a scenario, not a forecast; real audio duty and thermal throttling must be logged.

Mode	Power hypothesis	Duty in example	Energy / hour
Active conversation	12.0 W	20%	2.40 Wh
Light tool / speech	5.0 W	10%	0.50 Wh
Listening / idle	0.7 W	70%	0.49 Wh
Total	-	100%	3.39 Wh
45 Wh x 90% usable / total	-	-	11.9 h scenario

The enclosure must be modeled as a transient thermal system. Stage 1 records wall power and temperature; Stage 2 adds rail telemetry and FPGA die temperature. A '15 W device' is meaningless without ambient, skin-contact geometry, airflow, burst duration and throttle policy.



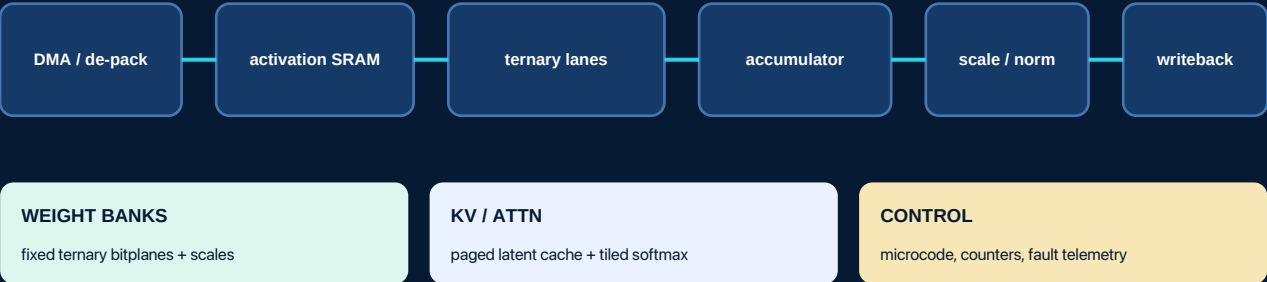
FPGA mapping

Use reconfigurable logic to falsify the accelerator thesis cheaply

STAGE 2 CORE

Instrument one repeatable kernel before mapping a whole model

Candidate FPGA decode dataflow - one layer at a time



Optimization hypothesis: replace general multiply paths with sign-select + zero-skip, while retaining mixed-precision islands.

Verification requirement: bit-exact golden vectors at each boundary; counters must expose stalls, sparsity and bytes moved.

The first RTL target is a representative transformer linear layer with the exact block shape, ternary packing, scales and activation precision selected in Stage 1. Host software emits golden input/output tensors. The FPGA reports cycles, useful lane operations, zero skips, bank conflicts, DMA bytes, stalls and error counters. Only after bit-exact closure should attention and full-layer scheduling be added.

ROOFLINE METHOD

For each kernel, arithmetic intensity $I = \text{useful operations} / \text{bytes transferred}$. Attainable throughput is $\min(P_{\text{peak}}, I \times BW_{\text{sustained}})$. Report both dense-equivalent operations and physical add/sub operations; otherwise zero-skipping can make comparisons misleading. Use sustained measured bandwidth, not memory data-sheet peak.

Kernel milestone	Proof artifact	Pass gate	Why it matters
Ternary GEMV tile	RTL + golden vectors	bit-exact; no unexplained overflow	Numerical foundation
Packed weight DMA	trace + counters	>80% of measured link BW on long burst	Validates packaging
Double-buffered layer	cycle profile	compute and DMA overlap visible	Exposes bottleneck
Tiled attention	reference trace	bounded SRAM; stable softmax error	Long-context path
Decode loop slice	tokens + power trace	>2x joules/token vs matched software baseline	System-level reason to continue

A fixed model enables compile-time bank assignment, address generation, pruned zero routing and protected-weight sidecars. But hard-wiring too early creates a brittle accelerator. Stage 2 should keep scales, shapes, precision modes and microcode configurable enough to run ablations. The goal is to learn what deserves to become fixed in an ASIC.

FPGA resource closure must include timing at target voltage, BRAM/URAM banking, DSP usage for scales and norms, routing congestion, host-transfer limits, and toolchain reproducibility. A post-synthesis TOPS estimate is not a product result.



Safety, privacy and control

A local model reduces disclosure; it does not remove authorization risk

NON-NEGOTIABLE BOUNDARY

The neural model proposes. Verified software decides.

Android accessibility services can inspect interface content and act on behalf of users, which makes them useful and security-sensitive [9]. The product must prefer structured application APIs, use accessibility only under explicit enablement, and require a fresh confirmation for high-impact transitions. Android’s hardware-backed Keystore can keep key material outside the app process when supported [8].

Risk class	Examples	Default policy	Required evidence
R0 observe	battery, current screen title	allow within granted scope	audit record
R1 reversible	open app, scroll, draft text	allow with visible indicator	postcondition observed
R2 communicative	send message, share file	preview + explicit confirmation	destination and content bound
R3 consequential	payment, delete, unlock, location share	strong user presence; narrow API	biometric / physical confirmation
R4 prohibited	disable safeguards, silent credential export	deny	tamper-evident denial log

CONFUSED-DEPUTY DEFENSE

Tool results, webpages, messages and screenshots are untrusted input. They cannot redefine policy or grant capabilities. The model sees capability handles scoped to a task, not bearer credentials. The policy gate canonicalizes arguments, binds confirmations to an exact action digest, expires them quickly, rate-limits repetitive attempts and verifies postconditions.

PRIVACY DATA FLOW

Data	Default location	Retention	User control
Raw microphone frames	ring buffer	seconds unless explicitly recorded	hard mute + indicator
Transcript	encrypted local	configurable	inspect / delete / export
KV cache	volatile memory	session / pressure eviction	clear conversation
Long-term memory	encrypted database	until deleted or expired	item-level controls
Tool credentials	hardware-backed key store	until revoked	per-connector revoke
Audit log	encrypted local	bounded rotation	inspect / export / purge policy

Local inference provides a meaningful privacy advantage only if telemetry, crash dumps, optional cloud escalation and companion backups obey the same disclosure model. 'On-device' must be verified with network-off tests and an egress inventory.



Stage 1 - software proof

Freeze measurement before freezing silicon

0-12 WEEKS

Build the evidence-generating reference system

Stage 1 runs on the Mac and, where useful, an embedded target. It does not pretend to be the future puck. It establishes a reproducible local agent, evaluates candidate models and attention/quantization variants, and captures traces that size the Stage 2 accelerator.

Work package	Weeks	Deliverable	Exit gate
S1.1 harness	1-2	versioned prompts, audio, actions, telemetry schema	one-command repeatable run
S1.2 model census	2-4	FP16/4-bit/ternary-native candidate scorecard	quality and license shortlist
S1.3 streaming voice	3-6	endpoint-to-audio trace and interruption handling	<450 ms P50 nominal
S1.4 memory system	4-8	KV bytes/token, retrieval, eviction tests	32K nominal within budget
S1.5 action controller	4-9	typed schemas, policy, confirmations, replay suite	zero unconfirmed R2/R3 in suite
S1.6 co-design trace	8-12	layer shapes, histograms, traffic and golden tensors	Stage 2 kernel frozen

MODEL SCORECARD

Dimension	Metric	Report
Language quality	perplexity + public task suite + targeted agent tasks	mean, bootstrap CI, per-task delta
Quantization loss	delta against same checkpoint / tokenizer	layer sensitivity and protected fraction
Speech interaction	endpoint-to-first-audio; interruption success	P50/P95/P99 by noise and prompt
Tool reliability	proposal validity; execution success; unsafe attempt rate	risk-class confusion matrix
Memory	resident bytes; KV bytes/token; page faults	context-length curves
Energy	joules/turn and joules/generated token	wall meter, sampling rate, ambient
Reproducibility	hashes, seeds, software and firmware versions	machine-readable run manifest

Stage 1 go/no-go: continue if at least one legally usable model meets the agreed quality floor after the selected low-bit treatment, the software agent meets the latency and action-safety gates, and measured traces support a credible FPGA bandwidth plan. Otherwise revise model size, context target or product envelope before hardware work.



Stage 2 - FPGA proof

Prove measured energy and latency advantage on representative workloads

3-9 MONTHS

A sequence of increasingly complete hardware experiments

Phase	Hardware scope	Measurement	Gate
A	one ternary GEMV tile	frequency, accuracy, lane utilization	bit-exact and timing-clean
B	full linear layer + DMA	sustained bandwidth, stalls, joules	measured roofline explained
C	attention/KV tile	bytes/token, softmax error, context scaling	bounded memory and quality
D	one transformer block	end-to-end block latency and energy	matches reference within tolerance
E	decode-loop slice	TTFT component, tok/s, joules/token	>2x energy benefit vs matched baseline
F	multi-block or emulation	thermal soak, fault recovery, tool reproducibility	ASIC evidence package

COMPARISON CONTRACT

The FPGA baseline and software baseline must use identical tensors, sequence lengths, batch size, numerical target and measured system boundary. Report FPGA board power both idle-subtracted and absolute. Include host energy when the host performs preprocessing or fallback kernels. Separate prefill and decode.

STAGE 2 EXIT PACKAGE

Artifact	Minimum content
Model manifest	architecture, shapes, precision map, checkpoint and package hash
Numerical report	golden vector agreement, task delta, overflow and saturation
Performance model	roofline assumptions reconciled with hardware counters
Power / thermal	rail and board power, ambient, duty cycle, throttling
RTL quality	lint, CDC, timing, utilization, deterministic build container
Product projection	explicit scaling from prototype node and memory to puck; sensitivity ranges
ASIC gate	ranked fixed vs configurable features, NRE and schedule risk

No ASIC sign-off occurs because an FPGA demo 'works.' Sign-off requires a stable trained model, stable decoder, stable package ABI, a measured bottleneck model, a software fallback path, and economics that survive sensitivity analysis.

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Validation and falsification

Benchmarks are testable product requirements, not demo choreography

EVIDENCE PYRAMID

Every headline metric needs a trace behind it

Layer	Question	Evidence
Unit	Is each numerical primitive correct?	golden vectors, property tests, saturation sweeps
Kernel	Does the mapped operation close?	cycles, bytes, stalls, power, bit accuracy
Model	Does quality survive architecture and precision?	paired evaluations and ablations
Agent	Can it plan and execute approved work?	scenario suite with postcondition scoring
Interaction	Does it feel responsive and interruptible?	timed audio trials in noise strata
Device	Can it sustain the workload safely?	thermal soak, battery scenario, fault injection

REQUIRED ABLATIONS

Ablation	Hold constant	Learn
FP16 vs 4-bit vs ternary	checkpoint family, tokenizer, eval set	quality cost of precision
MHA/GQA/latent KV	parameter budget and training data as feasible	memory-quality frontier
No speculation vs speculation	target model and sampling distribution	accepted tokens/step and true latency
No retrieval vs retrieval	active-context budget	memory utility and contamination risk
Model-only vs policy gate	same tool scenarios	safety contribution of deterministic layer
Software vs FPGA kernel	exact tensor workload	energy and latency advantage

STATISTICS

Pre-register primary metrics and thresholds. Use paired examples for model comparisons; report confidence intervals, not only means. For latency use quantiles and sample counts. For agent actions, classify both false execution and false refusal by risk class. Retain failure transcripts with secrets redacted. Never tune against a hidden final test set.

FALSIFIERS

Hypothesis	Evidence that falsifies it	Response
Ternary 7B-9B retains required quality	persistent task loss beyond agreed margin after native training	use higher precision / smaller scope
Pocket bandwidth supports real-time decode	measured bytes/token exceeds sustainable system bandwidth	reduce size, increase reuse, alter memory
Compressed context preserves utility	long-context retrieval / recall fails target	shorter hot cache + external memory
FPGA maps efficiently	routing, timing or memory stalls erase energy benefit	redesign dataflow or stop hardware path
Voice feels real-time	P95 first audio remains above gate	smaller model, speculative path, interaction redesign

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Risk register and program decisions

The largest risks are coupled, so gates must be cross-functional

PRE-MITIGATION VIEW

Quality, bandwidth and authorization dominate



Risk	Early indicator	Mitigation	Owner artifact
Ternary quality gap	outlier-heavy layers; tool JSON regressions	native training, mixed-precision islands	quantization sensitivity map
Weight bandwidth	decode tracks bytes, not compute	packing, stacking, smaller active model	roofline reconciliation
KV growth	resident memory linear and too steep	GQA/latent design, hot/cold policy	bytes/token curve
Thermal discomfort	surface trend crosses envelope	duty scheduling, enclosure, lower active W	thermal transient model
Unsafe automation	proposal bypass / confirmation race	typed gate, capability scope, replay tests	risk-class scenario suite
Frozen obsolescence	tool schemas or knowledge drift	update adapters and retrieval, version ABI	compatibility matrix

PROGRAM PRINCIPLES

1. Freeze interfaces before implementations. 2. Measure system boundaries, not isolated kernels. 3. Keep immutable model identity separate from mutable memory and tools. 4. Prefer deterministic enforcement for authorization. 5. Make every product number traceable to either a source, an equation or a run artifact. 6. Treat negative results as stage success when they prevent premature silicon.

The strategically valuable output of the first year is not only a device demo. It is a co-design evidence base: an evaluated low-bit checkpoint, a stable package ABI, execution traces, a measured memory model, an action-safety architecture, and FPGA results that constrain an ASIC.

11

Reference configuration

A concrete starting point, deliberately labeled as provisional

DESIGN-SPACE ANCHOR

Prototype around a narrow, measurable configuration

Subsystem	Stage 1 reference	Stage 2 hypothesis	Product aspiration
Language core	1B-3B + selected 7B baseline	ternary representative layers	natively ternary 7B-9B
Weights	4-bit baseline + ternary experiment	2-bit physical + block scales	2-4 GB full package
Attention	GQA baseline	tiled GQA then latent experiment	trained compressed KV
Context	8K/32K test points	32K trace target	32K normal; 120K research
Working memory	host RAM instrumented	board + host explicit	8-16 GB LPDDR
Speech	streaming ASR / TTS baseline	selected encoder/decoder kernels	integrated low-latency voice
Vision	on-demand frame	encoder trace only	event-driven
Actions	local typed simulator	same controller	Android / desktop companions
Power	wall-meter baseline	rail + board telemetry	8-15 W active target

WHY NOT START AT 9B ON FPGA?

A full 9B mapping combines model-quality uncertainty, packing, board memory, host links, attention, runtime and toolchain closure. A representative slice can test the unique thesis - ternary fixed-layout efficiency - without confusing it with integration work. Scale only when the measured bottleneck model predicts the next step.

OPEN RESEARCH QUESTIONS

Question	Experiment
How sparse are trained ternary zeros by layer and token?	export histograms; compare static and dynamic skip utility
Which weights require protected precision?	layer-wise and channel-wise sidecar ablation
Can a latent KV variant train within the quality budget?	matched-data architecture study; cache reconstruction analysis
Does speculative decoding help at batch one?	sweep drafter size, acceptance and memory traffic [7]
Can a single-voice decoder reduce startup latency?	distill fixed speaker; measure intelligibility and prosody
What should be hard-wired?	compare configuration use across validated model variants

The target specification remains a research aspiration until both stages pass. The public project should publish failures and changed assumptions alongside successes.

12

Conclusion

The frozen-model idea is valuable because it creates constraints that can be measured

A DISCIPLINED PATH TO REAL-TIME

Build the assistant first; earn the chip

Pocket Pet AI is an ambitious but coherent co-design program. Freezing a model can stabilize weight placement, execution order, memory formats and personality identity. Ternary weights can materially reduce storage and replace many multiplies with simpler operations. Compressed attention, streaming speech and tool adapters can make a local companion more capable than a standalone chatbot.

None of those advantages eliminates physics or safety. Each generated token still moves information; context still consumes memory; latency spans the entire audio path; a pocket enclosure must reject heat; an agent that controls devices needs deterministic authorization. The whitepaper therefore recommends two explicit stages.

Stage	Purpose	Definition of done
Stage 1	Prove the local agent and create trustworthy traces	quality, latency, memory, energy and action gates pass on a reproducible software system
Stage 2	Falsify or validate model-specific acceleration	bit-exact FPGA slices show measured system-level energy and latency value, then scale

The project should publish a living assumptions ledger and update every target with measured evidence. If the final product is possible, this process will reveal the architecture. If a premise fails, it will fail early enough to change course. That is the fastest credible route from a beautiful concept to real, personal, local intelligence.

Recommended immediate action: freeze the Stage 1 benchmark contract before choosing the final model. The benchmark is the instrument panel for every model, memory and hardware decision that follows.

References and calculation notes

Primary papers and platform documentation used in this draft

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CALCULATION NOTES

Item	Equation / assumption	Result
9B ideal ternary payload	$9e9 \times \log_2(3) / 8 / 1e9$	1.783 GB decimal
9B physical 2-bit payload	$9e9 \times 2 / 8 / 1e9$	2.250 GB decimal
GQA KV bytes/token	$32 \text{ layers} \times 2 \times 8 \text{ heads} \times 128 \text{ dim} \times 2 \text{ B}$	$131,072 \text{ B} = 128 \text{ KiB}$
GQA KV at 32K	$131,072 \times 32,768 / 2^{30}$	4.0 GiB
GQA KV at 120K	$131,072 \times 120,000 / 2^{30}$	14.65 GiB (rounded 15.0 in prose)
Weight bandwidth at 20 tok/s	$2.25 \text{ GB} \times 20$	45 GB/s lower bound
Example duty energy	$12 \text{ W} \times .2 \text{ h} + 5 \text{ W} \times .1 \text{ h} + .7 \text{ W} \times .7 \text{ h}$	3.39 Wh per mixed hour

All uncited product values are targets, hypotheses or derived examples. GB uses decimal units for package and bandwidth calculations; GiB uses binary units for KV capacity. Energy and battery examples exclude aging unless explicitly stated. No claim in this draft represents a fabricated prototype result.

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